



MPS Monitor SDK

MPS Monitor SDK & Simulator
API version: v3
Updated November 2023

Documentation references



API swagger:

<https://api.mpsmonitor.com/api3>

SDK download:

<https://cdn2.mpsmonitor.com/downloads/MpsMonitor.Simulator.zip>

MPS Monitor User's Guide:

<https://portal.mpsmonitor.com/doc/3DLSUQ2P>

Main entities



DEALER: the dealer is the highest-level entity in the application. It defines the perimeter for all other entities and for standard settings like alert levels, supplies, notifications and exports.

CUSTOMER: customers are the business entities in which devices are present. A customer belongs to a dealer. You need to create a customer in order to install the DCA and add devices to MPS Monitor.

DCA CONNECTOR: the DCA is the agent installed in the customer, to collect data from devices. In MPS Monitor, it is referred as a connector.

DEVICE: devices are printers installed into a customer and monitored by a connector. Devices are classified as Managed or Unmanaged; Unmanaged devices' data cannot be accessed via API.

Consumables management entities



LEVEL: consumable levels are updated at every data collection and stored in the database

ALERT: an alert is an event that generates the request for a consumable. It is triggered by an automatic or manual event and needs to be managed; an automatic alert is created when the level of a consumable goes below a defined threshold.

CONSUMABLE SET: a consumable sets is a list of consumable Part Numbers associated to one or many devices, that will be used to create shipments.

SHIPMENT: the shipment is the event that is created to “manage” the alert: as the alert is the need for a consumable, the shipment is the fulfillment of that need. To create a shipment, an associated consumable Part Number needs to be available.

➤ [See chapter 4 and 5 of the MPS Monitor User’s Guide for details on consumables management process](#)

Meters management entities



STANDARD COUNTER: Mono and Color total counters are collected and stored in the database on a daily basis; all the history of standard counters for a device is kept in the database until the device is present and classified as Managed.

DETAILED COUNTER: all the format and function counters available on each device are collected daily, but only the last updated value for each counter is stored. Detailed counters are available on the Portal and via APIs as a key – value table.

BLENDED COUNTER: to allow an easy use of detailed counters, they can be aggregated or renamed to create blended counters, which are available as columns in reports. The whole history of blended counters is kept in the database.

➤ [See chapter 7.6.2 and 9.13 of the MPS Monitor User's Guide for details on Meters management](#)

Customer entities



OFFICE: offices are intended as the physical locations where devices are present. They can be associated to physical and email addresses, and also to IP subnetworks, to enable automatic association of devices to offices.

PROJECT: projects allow to group devices into several smaller groups into a customer, or to identify devices in different customers that are linked to a specific group.

[See chapter 3.7 and 3.8 of the MPS Monitor User's Guide for details on Offices and Projects management](#)

Where to start?



If you are not an existing MPS Monitor user:

1. Go to <https://www.mpsmonitor.com/> and click on Request a Demo, then fill the form with your company and personal data
2. Log in to the MPS Monitor Portal with the credentials provided
3. Access the Developer menu in the MPS Monitor Portal
4. Create the API Credentials in order to access our API's
5. Create a new API user to use in your integration application
6. Access the REST API documentation at this URL <https://api.abassetmanagement.com/api3>
7. Download our .NET SDK and Simulator to start the integration

Simulator - Authentication



- Fill the fields with your API Client and API user credentials and choose the right Stack to use in the EndPoint Dropdown.
- * (To access the staging stack you need to send a request to help@mpsmonitor.com with your public internet address and the username of the user that you want to use to test your integration)

Selected elements

Dealer: Remove Dealer Customer: Remove Customer Device: Remove Device

Auth Dealer Customer Devices Supplies HP SDS Service Request

Authentication

EndPoint	ClientId	ClientSecret	UserName	Password	
Prod					Auth

<http://localhost:34287/>

Response Request

Overview of the Simulator



- “Selected elements” area allows you to see the filter applied to the data.
- With the tabs you can navigate through the different type of object available on MPS Monitor
- In the “Request and Response” area you can see which parameters are used to call the API’s and the JSON’s response returned for any API call.

A screenshot of the MPS Monitor Simulator web interface. The interface has a top navigation bar with tabs: Auth, Dealer, Customer, Devices, Supplies, and HP SDS Service Request. The 'Dealer' tab is currently selected. Below the navigation bar, there are three buttons: 'Remove Dealer', 'Remove Customer', and 'Remove Device'. The main content area is divided into sections. The first section is titled 'Dealers (Max. 20)' and contains a 'Get Dealers' button, a 'Selected Dealer' dropdown menu, and an 'Id' input field. Below this is a 'Device' section with tabs for 'Alert', 'Deliveries', and 'Offices'. The 'Alert' tab is selected. Below the tabs is a 'Pagination' section with a 'Get Devices' button. The bottom section is titled 'Devices' and contains a 'Response' and 'Request' tab. The 'Response' tab is selected, and it displays the text 'User authenticate status: True'. Red rectangular boxes highlight the navigation tabs, the 'Remove Dealer' button, and the 'Response' tab.

USE CASE 1: Reading meters



Extract the Counters with [/Counter/List](#)

Available filters: CustomerCode, Serial or AssetNumber, Date Range (max 1 month)

Extract the last Counters Detailed with [/Counter/ListDetailed](#)

Available filters: CustomerCode, Serial or AssetNumber

Extract the Blended counters with [/Counter/ListBlended](#)

Available filters: CustomerCode, Serial or AssetNumber, Date Range (max 1 month)

USE CASE 2: Reading device data with levels



Extract the last levels of all device's consumables with [/Consumable/List](#)

Available filters: CustomerCode, Serial or AssetNumber

Extract the list of devices with [/Device/List](#)

Available filters: CustomerCode, Serial or AssetNumber, Managed/Not managed status

Extract a specific device data with [/Device/Get](#)

USE CASE 3: Getting Alerts



In MPS Monitor an alert has 2 main status “Open” and “Closed”. These 2 main status are divided into other specific statuses:

- **“Open” alerts:**
 - Not Managed (indicates that the alert was open at a certain time in the system).
 - A not managed alert can be set as “Hidden” in the system and it means “I saw the alert, but I don’t want to manage it, so I don’t want to see it in the list of alerts to manage”
- **“Closed” alerts:**
 - Managed (indicates that a consumable was associated to the alert). Shipment data and logistic process can be added during this action.
 - Installed (indicates that a new consumable was installed in the printer).
 - Cancelled (indicated that the alert was canceled by operator)

Get the alerts list with: [/SupplyAlert/List](#)

Get only the “Open” alerts with these filters: "ManageOption": “NotManaged”, "HiddenOption": “NotHidden”

Get a single alert with: [/SupplyAlert/Get](#)

USE CASE 4: Manage an Alert



To manage an alert, you need to specify a Part Number. The part number should be present in the DEALER consumable list or you will need to specify a new consumable object in the alert manage API request.

Get the available consumables for a device with:

[/SupplyAlert/GetAvailableSuppliesForADevice](#), this call will return an object with:

- AvailableSupplies: The consumables provided by the CONSUMABLE SET associated to the printer
- CurrentSupplies: A matching consumable with the CONSUMABLE LIST and the part number read from the printer (were available)
- LastUsedSupply: Last shipped consumable from same alert type and color
- SuggestedSupplies: Suggested consumable from internal consideration

Manage an alert with [ShippedSupply/CreateOnAlert](#)

- By specify a partnumber "PartNumber": "CE250XC"

USE CASE 5: Update a device

There are 3 different API calls for updating a device:

- Update the full device data with /Device/Update
- This API updates all the DEVICE fields
- Update the CONSUMABLE SET associated with
[/Device/UpdateSupplySetAssociation](#)
- Change the MANAGED/NOT MANAGED status with: [/Device/Manage](#)

USE CASE 6: Connectors and Configurations list



- Get the CONNECTOR list with: [/Explorer/GetConnectors](#)
 - With filters: by Customer, if Clustered or not, by Last communication, with or without configuration
- Get the CONNECTOR CONFIGURATIONs associated to a CONNECTOR with: [/Explorer/Configuration/List](#)
- Get the single CONNECTOR CONFIGURATION with: [/Explorer/Configuration/Get](#)

USE CASE 7: Set Alert Thresholds



The default alert threshold value for any consumable is “level = 10”.

You can specify other values and thresholds for toners and maintenance kits at DEALER, CUSTOMER and DEVICE level; our alert level system will search in this order the value to compare with the level retrieved from the printer.

To change levels in the different objects use these API calls:

- [/AlertLimit2/Dealer/UpdateDefault](#) and [/AlertLimit2/Dealer/UpdateProduct](#)
- [/AlertLimit2/Customer/UpdateDefault](#) and [/AlertLimit2/Customer/UpdateProduct](#)
- [/AlertLimit2/Device/UpdateDefault](#)

USE CASE 8: Create Customer

Create a new CUSTOMER with: [/Customer/Create](#)

- You just need to specify a description of the customer, the DEALER and the TIMEZONE where the CUSTOMER belong to
- By setting the customer email address and activating the flag “SendExplorerInstallationInvitation” you can send the installation invitation.
- You can download the DCA Setup link using the [/Explorer/GetExplorerSetupLink](#) or in the answer of the [/Customer/Create](#)

USE CASE 9: Getting Invoices

Invoices from Dealer to Customers can be computed by MPS Monitor based on Contracts and Invoice Plans, and are available by means of the following APIs

- Extract the list of invoices in a period with [/billing/GetCustomersInvoicesList](#)
 - Available filters: IssuerCode, FromIssueDate, ToIssueDate
- Extract invoice details with [/billing/GetCustomerInvoice](#)
 - Available filters: Id

USE CASE 10: Custom result using a report



A user can build the report using Analytics/Export feature, save it and then call an API to get the report data.

This can improve and simplify communication and API usage. A dealer can get combined data in a single API call instead of calling several standards API and combine the result client-side.

Reports built by users are available by means of the following APIs

- Extract the result of a report in Excel format with [/analytics/GetReportFileResult](#)
 - Available filters: IdReport, ReportFormat (excel, csv)
- Extract the result of a report in JSON format with [/analytics/GetReportResult](#)
 - Available filters: IdReport



mpsMonitor
Printer Monitoring in the Cloud

ANALYTICS

For more information, please log a ticket in the Portal or contact help@mpsmonitor.com